SHIV NADAR UNIVERSITY CHENNAI

(State Private University Established Under the Shiv Nadar University Act, 2018)

School of Engineering

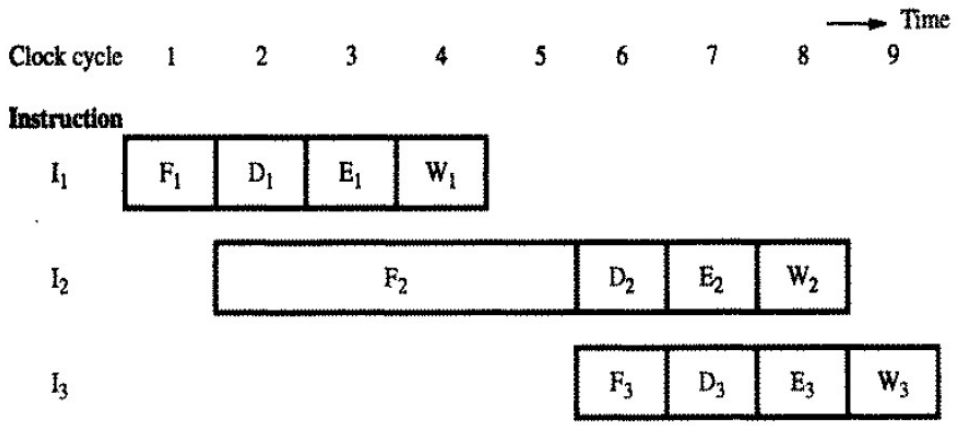
Department of Computer Science and Engineering

End Semester Examinations 2021-2022 Even

Question Paper

Name of the Programme: Common to B.Tech. AI & DS and B.Tech. CSE (IoT)		Semester: **II**
Course Code & Name: **CS1004 COMPUTER ORGANIZATION AND ARCHITECTURE**		
Regulations: **Regulation 2021**		
Time: **3 Hours**		Maximum: **100 Marks**

Q.No		Questions	Marks	CO#	KL#
Part-A - Answer all the questions (5 × 2 = 10 Marks)					
1	a	The idea of cache memory is based i. On the property of locality of reference ii. On the fact that references generally tend to cluster iii. On the heuristic 90-10 rule iv. All the above	2	CO1	KL1
2	a	Computers use addressing mode techniques for i. giving programming versatility to the user by providing facilities as pointers to memory counters for loop control ii. to reduce no. of bits in the field of instruction iii. specifying rules for modifying or interpreting address field of the instruction iv. All the above	2	CO1	KL2
3	a	If the Internal circuit of a 128-bit memory contains 14 external connections out of which 4 are address lines, how many data lines are there? i. 7 ii. 8 iii. 9 iv. 10	2	CO2	KL4
4	a	If the main memory of 128 blocks and the cache memory of 32 blocks uses direct mapping, then how many bits are allocated for tag? i. 3 bits ii. 4 bits iii. 5 bits iv. 6 bits	2	CO2	KL4
5		The communication between the components in a microcomputer takes place via the address and i. I/O bus ii. Data bus iii. Address bus iv. Control bus	2	CO3	KL2
Part-B - Answer any 5 questions (5 × 10 = 50 Marks)					
6	a	What is the Basic Performance Equation? How do the different parameters in the equation affect the performance of RISC and CISC systems?	5	CO1	KL2

	b	Discuss the effect of pipelining and superscalar operation on the performance of the processor.	5	CO1	KL5
7	a	What is cache mapping? Discuss the different mapping techniques assuming a main memory consisting of **m** blocks and cache having **n** blocks.	10	CO2	KL3
8	a	How can a 1K × 1 memory chip be organized? Discuss in detail.	5	CO2	KL5
	b	Discuss in detail the internal organization of a 2M × 8 dynamic memory chip.	5	CO2	KL5
9	a	Discuss the advantage of multi-bus over single-bus datapath organization inside a processor.	5	CO5	KL2
	b	Discuss different types of ROMs in detail.	5	CO3	KL2
10	a	The following figure shows an instruction being delayed as a result of cache miss. Redraw this figure assuming the pipeline fetch unit contains an instruction queue that can hold up to four instructions and that the instruction fetch unit reads two instructions at a time from the cache. 	5	CO4	KL4
	b	Consider a n-stage pipeline. Discuss different factors that affect the performance of this pipeline.	5	CO4	KL3
11	a	What is Superscalar Operation in pipelining? Explain how multiple instructions can be executed concurrently using Superscalar Operation.	10	CO3	KL3
Part-C - Answer any 2 questions (2 × 20 = 40 Marks)					
12	a	Explain the single-bus organization of a datapath inside a processor in detail. Assume the processor consists of the following components: i. Registers: IR, PC, MAR, MDR, n-general purpose registers, 3 temporary registers ii. ALU unit iii. Control unit	10	CO4	KL3
	b	Explain how registers transfer data between them while located inside a processor. Illustrate with a step-by-step example.	10	CO5	KL3

13	a	Explain the effect of Data hazard in a pipeline with an example. What modifications can we do to the datapath in a processor to handle Data hazard?	10	CO4	KL3

	b	Consider the following sequence of instructions:
		Add #20,R0,R1
		Mul #3,R2,R3
		And #$3A,R2,R4
		Add R0,R2,R5
		In all instructions, the destination operand is given last. Initially, registers R0 and R2 contain 2000 and 50, respectively. These instructions are executed in a computer that has a four-stage pipeline (Fetch, Decode, Execute, Write). Assume that the first instruction is fetched in clock cycle 1, and that instruction fetch requires only one clock cycle.
		i. Draw the pipeline diagram for the above set of instructions. Describe the operations performed by each pipeline stage during each of clock cycles 1 through 4.
		ii. Give the contents of the interstage buffers B1, B2 and B3 during clock cycles 2 to 5.
		Note: The symbol $ represents hexadecimal value. Assume all registers store data in binary form.
			10	CO5	KL5

14	a	Explain how Instruction hazard affects a pipeline with an example. Discuss how one can handle Instruction hazard by modifying the hardware organization of the CPU.	10	CO4	KL2
	b	Discuss the memory hierarchy of a computer system in detail.	10	CO5	KL2

KL – Bloom's Taxonomy Levels

(KL1: Remembering, KL2: Understanding, KL3: Applying, KL4: Analyzing, KL5: Evaluating, KL6: Creating)

CO – Course Outcomes